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Strategic Decision Dynamics in Live Streaming Supply Chain: An Evolutionary Game Approach to Human-Virtual Agent Competition

Jingyi Zhang*, Ruiqi Yue

School of Economics and Management, Tianjin University of Technology and Education, 300222 Tianjin, China

* Correspondence: Jingyi Zhang (15838977887@163.com)

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Abstract: Rapid expansion of e-commerce live streaming has introduced new strategic choices regarding the use of human or virtual agents within the platform-based ecosystem. However, the decision dynamics underlying such choices remain insufficiently understood, particularly in the presence of multiple interacting stakeholders. This study developed an evolutionary game model to analyze the strategic interactions among brands, streaming platforms, and consumers. The framework incorporated agent heterogeneity in terms of consumer attraction, information transmission, and trust formation, while explicitly modeling cost structures and revenue-sharing mechanisms. Replicator dynamics were derived to characterize strategy evolution, and system stability was examined through Jacobian analysis. The results demonstrated that the revenue-sharing coefficient critically determined system trajectories. Higher sharing ratios led to convergence toward human agents and customized services, whereas lower ratios promoted virtual agents and standardized solutions. The findings further revealed that equilibrium outcomes were jointly shaped by cost–benefit configurations, consumers' responses, and platform incentives. This study provided an analytical foundation for comprehending strategy formation in digital commerce systems and contribute to the design of incentive mechanisms in a platform-mediated environment.

Keywords: Evolutionary game theory; Big data analytics; Jacobian; Strategic interaction; Heterogeneity

1 Introduction

China is actively advancing the development of new quality productivity based on emerging technologies, in order to transform traditional human and resource-driven economic growth models to enhance production efficiency and quality while creating new value and economic growth drivers [1]. Cutting-edge technologies such as artificial intelligence, big data, and virtual reality have laid a technical foundation for this productivity revolution, enabling the economy to break free from conventional growth patterns and achieve significant improvements in total factor productivity. As technologies like AI continue to evolve, e-commerce live streaming has also been impacted by technological advancement, with an increasing number of brands adopting computer-generated virtual digital hosts. Traditional e-commerce live streaming revolves around the core elements of "people, products, and venues", while virtual anchors, as a new form of productivity, have injected fresh vitality into the upgrade and high-quality development in the industry [2]. By leveraging computer-generated imagery (CGI), they have innovated and transcended the conventional "people-product-venue" model [3]. Firstly, virtual anchors provide round-the-clock uninterrupted services and enable personalized interactions with viewers during broadcasts, freeing up human resources. Secondly, their live streaming rooms have undergone technological innovations. These virtual studios can adapt to product characteristics, thus enhancing viewers' immersive experience and redefining consumers' perception of the "venue".

According to the 53rd "China Internet Network Development Status" report released by the China Internet Network Information Centre in March 2024, as of December 2023, the number of e-commerce live streaming users in China reached 597 million, an increase of 82.67 million compared with December 2022, accounting for 54.7% of the total internet users [4]. Data from iiMedia Research indicated that the core market size of China's virtual humans in 2023 was 20.52 billion yuan, and it is expected to have reached 48.06 billion yuan by 2025 [5], implying that virtual anchors still have significant potential of growth in the market. When compared with live streaming hosts with real faces, virtual hosts exhibit distinct differences in attracting consumers, delivering information, and building trust [6]. Current research on the underlying mechanisms through which virtual hosts influence consumer purchasing decisions remains in its exploratory phase, lacking systematic and in-depth studies. Both live streaming hosts and virtual hosts have their respective strengths and weaknesses. While live streaming services continue to improve and virtual hosting technologies evolve, brands face growing challenges in selecting suitable hosts.

2 Research Background

The 1992 science fiction "Snow Crash" mentions virtual humans and the metaverse, where real people will have their own virtual avatars in the "Metaverse". This virtual world is called the "metaverse", and virtual humans are also referred to as avatars (Avatars) [7]. The "China Virtual Digital Human Influence Index Report", jointly released by the State Key Laboratory of Media Convergence and Communication at China Media University and others, defined virtual digital humans as interactive virtual images that possessed the appearance, behavior, and thinking of "human beings". These beings were created through technologies such as computer graphics, biotechnology, motion capture, brain science, deep learning, and language synthesis [8]. Many scholars conducted comparative experiments between live streaming hosts with real faces and virtual hosts to study the differences in their impact on consumers' purchasing behavior. Some research indicated that virtual hosts did not demonstrate superior advantages over real hosts in attracting viewership and driving purchases. Wan and Jiang [9] found that consumers perceived virtual hosts as less warm, trustworthy, useful, and conversational, compared with real hosts in live streaming e-commerce. There was no evidence suggesting that virtual hosts were more enjoyable, user-friendly, or purchase-encouraging than their human counterparts. Lou et al. [10] argued that while virtual hosts positively influenced brand image building and awareness enhancement, their lack of authenticity and weak hyper-social connections with consumers resulted in diminished persuasive power for purchase stimulation. However, other scholars contended that virtual hosts might more effectively spark consumer curiosity through technical expertise and psychological needs, thereby attracting viewership and purchases. As revealed in a survey, Gerlich [11] compared virtual influencers and real hosts and concluded that virtual influencers enjoyed greater popularity out of factors such as professional knowledge, reliability, relevance, and consumer trust, as these significantly influenced positive reception and purchasing decisions. Wei et al. [12] conducted an intergroup experimental comparison involving virtual endorsers, real endorsers, and various advertising scenarios, to reveal that virtual endorsers exerted a more significantly positive influence on consumer purchase intent, compared with real endorsers by fulfilling consumers' psychological needs.

3 Construction of Evolutionary Game Model

3.1 Model Assumptions

In live streaming commerce, both brand owners and live streaming platforms operate as bounded rational actors. They are not perfectly rational decision-makers, as they cannot obtain complete decision-making information nor possess unlimited information-processing capabilities. Therefore, when making decisions, they do not pursue theoretically optimal solutions but tend to adopt satisfactory strategies that meet their own revenue requirements [13, 14]. Platforms provide brands with two types of hosts: Real-person anchors and virtual anchors, with the latter being computer-generated characters controllable by brands. Both parties engage in cost-benefit analyses. While human streamers possess advantages like on-site control capabilities unavailable to virtual anchors, they also face the risk of being replaced [15]. This study investigated revenue-sharing coefficients between brands and platforms to analyze the evolutionary game dynamics among brand owners, live streaming companies, and consumers in live streaming commerce. Therefore, the following hypotheses have been proposed:

Hypothesis 1: Tripartite strategy selection for brand owners. When selecting live streamers, they can choose between virtual and human streamers, with the probability of choosing human streamers denoted as x and virtual anchors as $1 - x$. For live streaming companies, they may offer customized or non-customized virtual/human streamers, with customization probabilities of y and non-customization probabilities of $1 - y$, respectively. Customized human streamers are exclusively assigned to promote specific brands, while non-customized anchors can promote multiple brands simultaneously. Virtual streamers can be customized through appearance, tone of voice, and behavioral patterns to align with brand products, or purchased directly from digital avatar systems using pre-designed models, resulting in homogeneous content lacking distinctiveness. Consumer behavior follows a purchase/non-purchase dichotomy, with buying probability z and non-purchasing probability $1 - z$.

Hypothesis 2: Brand decision-making assumptions. If brands choose live streaming hosts, they must share the revenue generated from live streams with the streaming company proportionally. Let the revenue-sharing coefficient for customized streamers be k_1 and that for non-customized hosts be k_2. Customized streamers receive higher revenue shares than non-customized ones due to greater brand alignment, hence satisfying the condition $0 < k_2 < k_1 < 1$. In addition, human streamers' salaries and commissions are fully covered by the streaming company, with brands focusing solely on revenue-sharing arrangements. When selecting virtual streamers, streaming companies provide digital avatar systems (costed C). The technological advantages of virtual streamers attract user engagement, thus generating additional brand traffic revenue R, while all live streaming revenue remains exclusively owned by the brand.

Hypothesis 3: Decision-making assumptions for live streaming companies. Live streaming platforms offer two types of hosts: Customized and non-customized. Customized live hosts require higher cultivation and development costs, with customized real hosts incurring additional training costs C_z and customized virtual hosts requiring higher cultivation costs C_v. Since customized streamers better align with brand image, the revenue generated by customized human streamers' live streams (pq_a) exceeds that of non-customized real streams (pq_b), where P and q_a q_b represent product prices and sales volumes during live broadcasts. Customized virtual streamers generate revenue R_c, while non-customized virtual streamers using pre-built templates from the system library face dual challenges: potential brand homogeneity leading to user visual fatigue, and rigid virtual streamers causing user attrition. The revenue for non-customized virtual streamers is R_n, resulting in $R_c > R_n$.

Hypothesis 4: Consumer Behavior Decision-making Assumption. It is hypothesized that consumers derive actual utility G from purchasing products. When shopping in live streaming sessions, consumers not only gain the product's intrinsic utility value but also experience shopping satisfaction. The study posited that the satisfaction levels differed between purchases from virtual streamers and human streamers: Consumers perceive satisfaction E_1 $(E_1 > 0)$ when buying from virtual streamers, while E_2 $(E_2 > 0)$ is observed when purchasing from human streamers. Model symbols and parameter definitions are detailed in Table 1.

Table 1. Description of the parameters

Sign	Meaning of the Parameters
x	Probability of brand owners choosing live streamers
y	Probability of human streamers choosing positive promotion
z	Probability of consumer purchase
k_1	Revenue-sharing ratio for live streaming companies when providing customized human streamers
k_2	Revenue-sharing coefficient for live streaming companies when using non-customized human streamers
p	Product price
q_a	Product sales volume of customized human streamer live rooms
q_b	Product sales volume of non-customized human streamer live rooms
C	Cost for brands to select virtual streamers
C_v	Additional cost for customized virtual streamers
C_z	Additional cost for customized human streamers
R_c	Customized virtual streamer live room revenue
R_n	Non-customized virtual streamer live room revenue
R	Traffic revenue brought to brands by consumers attracted to virtual streamers
G	Consumers' actual utility obtained from product purchase
E_1	Consumers' satisfaction derived from purchasing via virtual streamers
E_2	Consumers' satisfaction derived from purchasing via human streamers

3.2 Evolutionary Game Revenue Matrix

Based on model assumptions, a three-party evolutionary game payoff matrix was constructed for brand owners, live streaming companies, and consumers, as shown in Table 2.

Table 2. Revenue matrix of brand owners, streamers (human/virtual), and consumers

		Live Streaming Companies	Consumers: Purchase (z)	Consumers: Not Purchase $(1-z)$
Brand owners	Virtual streamers $(1-x)$	Customize (y)	$R_c - C + R$, $C - C_v$, $G + E_1$	$-C + R$, $C - C_v$, 0
		Not customize $(1-y)$	$R_n - C + R$, C, $G + E_1$	$-C + R$, C, 0
	Human streamers (x)	Customize (y)	$(1-k_1)\,pq_a$, $k_1 pq_a - C_z$, $G + E_2$	0, $-C_z$, 0
		Not customize $(1-y)$	$(1-k_2)\,pq_b$, $k_2 pq_b$, $G + E_2$	0, 0, 0

4 Stability and Evolution Path Analysis

4.1 Copy Dynamic Analysis

4.1.1 Replicator dynamics analysis of the brand

Let the expected payoff of the brand choose a virtual streamer be U_{11}, and the expected payoff of choosing a human streamer be U_{12}. According to evolutionary game theory, we have:

$$U_{11} = yz\left(R_c - C + R\right) + y(1-z)(-C+R) + (1-y)z\left(R_n - C + R\right) + (1-y)(1-z)(-C+R)$$
$$U_{12} = yz\left[\left(1-k_1\right)pq_a\right] + (1-y)z\left[\left(1-k_2\right)pq_b\right]$$

Then, the replicator dynamic equation of the brand is:

$$F(x) = x(1-x)\left(U_{11} - U_{12}\right) = x(x-1)\left[(C-R)(y-1)(z-1) - yz\left(R - C + R_c\right) + pq_b z\left(k_2 - 1\right)(y-1) - pq_a yz\left(k_1 - 1\right) + yz(C-R)(y-1)(z-1)\left(R - C + R_n\right)\right]$$

The first-order derivative is given by:

$$F'(x) = (2x-1)\left[(C-R)(y-1)(z-1) - yz\left(R - C + R_c\right) + pq_b z\left(k_2 - 1\right)(y-1) - pq_a yz\left(k_1 - 1\right) + yz(C-R)(y-1)(z-1)\left(R - C + R_n\right)\right]$$

4.1.2 Replicator dynamics analysis of the live streaming company

Let the expected payoff of the live streaming company which provides customized services be U_{21}, and the expected payoff of providing non-customized services be U_{22}. According to evolutionary game theory, we have:

$$U_{21} = (1-x)\left(C - C_v\right) + xz\left(k_1 pq_a - C_z\right) - x(1-z)C_z$$
$$U_{22} = (1-x)C + xz\left(k_2 pq_b\right)$$

Then, the replicator dynamic equation of the human streamers is:

$$G(y) = y(y-1)\left(C_v - C_v x + C_z x - k_1 pq_a xz + k_2 pq_b xz\right)$$

The first-order derivative is given by:

$$G'(y) = (2y-1)\left(C_v - C_v x + C_z x - k_1 pq_a xz + k_2 pq_b xz\right)$$

4.1.3 Replicator dynamics analysis of consumers

Let the expected payoff of consumers choosing to purchase be U_{31}, and the expected payoff of choosing not to purchase be U_{32}. According to evolutionary game theory, we have:

$$U_{31} = (1-x)\left(G + E_1\right) + x\left(G + E_2\right)$$
$$U_{32} = 0$$

Then, the replicator dynamic equation of consumers is:

$$H(z) = z(1-z)\left(U_{31} - U_{32}\right) = z(z-1)\left[(x-1)\left(E_1 + G\right) - x\left(E_2 + G\right)\right]$$

The first-order derivative is given by:

$$H'(z) = -(2z-1)\left(E_1 + G - E_1 x + E_2 x\right)$$

4.2 Stability Analysis of the Tripartite Evolutionary Game

4.2.1 Jacobian matrix

According to the replicator dynamic equations of the three parties, the Jacobian matrix J of the system can be derived,

$$J = \begin{bmatrix} \frac{\partial F(x)}{\partial x} & \frac{\partial F(x)}{\partial y} & \frac{\partial F(x)}{\partial z} \\ \frac{\partial G(y)}{\partial x} & \frac{\partial G(y)}{\partial y} & \frac{\partial G(y)}{\partial z} \\ \frac{\partial H(z)}{\partial x} & \frac{\partial H(z)}{\partial y} & \frac{\partial H(z)}{\partial z} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

where,

$$\left\{\begin{array}{c} a_{11}=(2x-1)\left[(C-R)(y-1)(z-1)-yz\left(R-C+R_c\right)+pq_bz\left(k_2-1\right)(y-1)-\right. \\ \left. pq_ayz\left(k_1-1\right)+yz(C-R)(y-1)(z-1)\left(R-C+R_n\right)\right] \\ a_{12}=x(x-1)\left[(C-R)(z-1)-z\left(R-C+R_c\right)-pq_az\left(k_1-1\right)+\right. \\ \left. pq_bz\left(k_2-1\right)+(2yz-z)(C-R)(z-1)\left(R-C+R_n\right)\right] \\ a_{13}=x(x-1)\left[(C-R)(y-1)-y\left(R-C+R_c\right)-pq_ay\left(k_1-1\right)+\right. \\ \left. pq_b\left(k_2-1\right)(y-1)+(2yz-y)(C-R)(y-1)\left(R-C+R_n\right)\right] \\ a_{21}=-y(y-1)\left(C_v-C_z+k_1pq_az-k_2pq_bz\right) \\ a_{22}=(2y-1)\left(C_v-C_vx+C_zx-k_1pq_axz+k_2pq_bxz\right) \\ a_{23}=-pxy\left(k_1q_a-k_2q_b\right)(y-1) \\ a_{31}=z\left(E_1-E_2\right)(z-1) \\ a_{32}=0 \\ a_{33}=-(2z-1)\left(E_1+G-E_1x+E_2x\right) \end{array}\right.$$

4.2.2 Equilibrium points

In asymmetric dynamic games, mixed-strategy equilibria are not evolutionarily stable equilibria [16]. Therefore, only pure-strategy equilibrium points of the evolutionary game system were analyzed. By setting $F(x)=0$, $G(y)=0$, and $H(z)=0$, eight pure-strategy equilibrium points in the evolutionary game system can be obtained. The stability analysis of specific equilibrium points is shown in Table 3.

Table 3. Stability analysis of equilibrium points

Equilibrium Points	Eigenvalues of Jacobian Matrix	Eigenvalue Sign	Stability
$E_1(0,0,0)$	$R-C,\ -C_v,\ E_1+G$	$(\times,-,+)$	No Stability
$E_2(1,0,0)$	$C-R,\ -C_z,\ E_2+G$	$(\times,-,+)$	No Stability
$E_3(0,1,0)$	$0,\ C_v,\ E_1+G$	$(\times,+,+)$	No Stability
$E_4(0,0,1)$	$-C_v,\ -E_1-G,\ pq_b(k_2-1)$	$(-,-,-)$	ESS①
$E_5(1,1,0)$	$0,\ C_z,\ E_2+G$	$(\times,+,+)$	No Stability
$E_6(1,0,1)$	$-E_2-G,\ k_1pq_a-C_z-k_2pq_b,\ pq_b(1-k_2)$	$(-,\times,+)$	No Stability
$E_7(0,1,1)$	$C_v,\ -E_1-G,\ R-C+R_c+pq_a(k_1-1)$	$(+,-,\times)$	No Stability
$E_8(1,1,1)$	$-E_2-G,\ C_z-k_1pq_a+k_2pq_b,\ C-R-R_c-pq_a(k_1-1)$	$(-,\times,\times)$	ESS②

Note: The symbol × denotes that relevant conditions need to be satisfied. At the equilibrium point, a serve as a reference index, while x, y, z represents the proportions of the three-party strategies, respectively. ESS = Evolutionarily Stable Strategy, a robust long-term equilibrium. ① $C_2-k_1pq_a+k_2pq_b<0, C-R-R_c-pq_a\left(k_l-1\right)<0$; ② $C_2-k_1pq_a+k_2pq_b<0, C-R-R_c-pq_a\left(k_1-1\right)<0$.

The table above calculates the eigenvalues of the Jacobian matrix for the equilibrium points of pure strategy combinations among three parties and determines their positive/negative signs. Under certain conditions, $E_4(0,0,1)$ and $E_8(1,1,1)$ are evolutionarily stable strategies, while other strategies exhibit instability due to at least one positive eigenvalue. The following sections will analyze these cases based on different scenarios.

Scenario 1: Benign Equilibrium State. If $C_z-k_lpq_a+k_2pq_b<0$ and $C-R-R_c-pq_a\left(k_l-1\right)<0$, then $E_8(1,1,1)$ is an evolutionarily stable equilibrium. In this case, the brand chooses human streamers for product promotion. Since customized human streamers feature higher brand relevance, they can generate greater payoffs for both the brand and the live streaming company. Therefore, the human streaming company provides customized human streamer services, and consumers choose to purchase, so as to obtain both emotional utility and practical product utility. Under this equilibrium, all three parties achieve positive payoffs, representing a benign equilibrium.

Scenario 2: Compromise Equilibrium State. The evolutionary stable equilibrium $E_4(0,0,1)$ indicates that brands opt for virtual streamers in live streaming, while streaming companies provide customizable virtual anchor services with consumer demand. Here, brands incur lower costs with virtual streamers compared with human streamers, leading to their adoption driven by profit motives. Streaming companies face higher costs in customization than in customization services, resulting in a preference for flexible virtual anchor solutions. This scenario arises from two factors: First, virtual streamers remain in development with consumer acceptance still under evaluation, and most streaming platforms lack mature virtual streamer technologies. Second, brands primarily use virtual streamers as pilot strategies in e-commerce live streaming, resulting in limited investment from both parties and a preference for customizable solutions. Last but not least, technical sophistication and novelty of virtual streamers create distinct experiential differences from human streamers during live commerce, thus stimulating purchase intent and ultimately achieving equilibrium.

5 Numerical Simulation

Based on previous analysis, the system can achieve an ideal evolutionary stable state characterized by human streamers, customization, and purchases. To explore how various factors influence strategic behaviors of stakeholders and facilitate system stabilization, numerical simulations were conducted using MATLAB R2023b software.

All simulation parameters in this study adopted dimensionless relative values rather than absolute monetary amounts, which was a standard approach in evolutionary game research. Combined with real data from live streaming e-commerce industry, public enterprise operational information, survey results, and existing mature literature, the relative magnitude of parameters was reasonably calibrated [17]. Specifically, additional cost for customized human streamers ($Cz = 18$) was slightly higher than that of virtual streamers ($Cv = 15$), while customized virtual streamer live room revenue ($Rc = 100$) exceeded non-customized revenue ($Rn = 80$), with all cost inputs far lower than total revenue. The actual industrial cost-benefit logic and practical conditions are shown in Table 4.

Table 4. Simulation parameter assignments

p	q_a	q_b	C	C_v	C_z	R_c	R_n	R	G	E_1	E_2
10	10	8	30	15	18	100	80	10	10	5	3

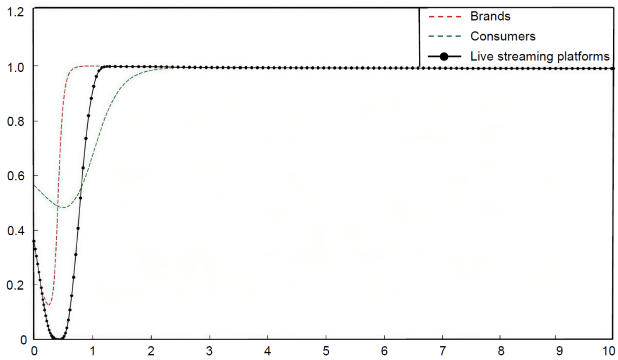

Figure 1. Impact of high partition coefficient on decision making by all parties

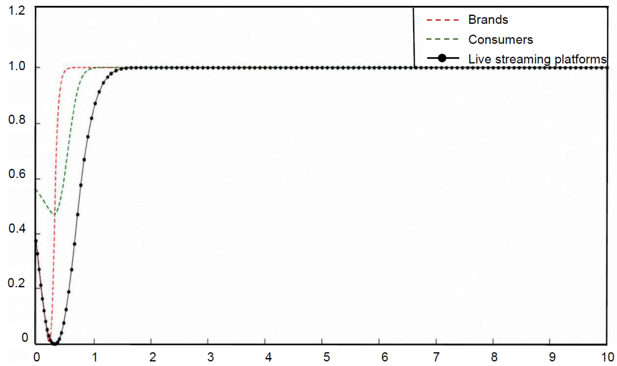

Figure 2. Impact of median division coefficient on decision making by all parties

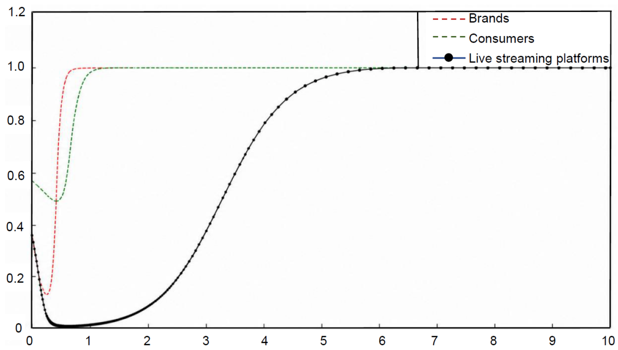

Figure 3. Impact of low allocation coefficient on decision making by all parties

Customization coefficients were set at 0.8, 0.5, and 0.2 for high, medium, and low levels respectively, while customization flexibility coefficients were configured at 0.6, 0.3, and 0.1. Given the disruptive impact of virtual streamers on e-commerce live streaming platforms and the resulting competitive pressure on human streamers, probabilities of initial strategy selection were defined as $x = 0.4$, $y = 0.5$, $z = 0.6$.

When brands grant higher revenue-sharing ratios to live streaming platforms, three key evolutionary trends emerge gradually. The probability that brands select human streamers, the probability that platforms provide customized services, and consumers' purchase intention all increase continuously and eventually stabilized at 1. The simulation results in Figures 1–3 depict the dynamic evolution of strategy selection probabilities over time. Among them, x denotes the probability that brands choose human streamers, y represents the probability that live streaming platforms provide customized services, and z indicates consumers' purchase probability. This result verifies that a high revenue-sharing mechanism could effectively motivate brands and platforms to adopt consumer-oriented strategies, optimize service quality, and enhance consumer experience, thereby forming a virtuous tripartite circulation.

5.1 Simulation Analysis of System Dynamic Evolution under High Coefficient of Partitioning ($k_1 = 0.8, k_2 = 0.6$)

Under high revenue-sharing coefficients, the strategic willingness of brands to select human streamers, platforms to deliver customized services, and consumers to make purchases all rise steadily and finally converged to 1. It indicates that sufficient revenue sharing could fully mobilize the initiative of all stakeholders and realize the optimal evolutionary stable state.

5.2 Simulation Analysis of System Dynamic Evolution under Equal Division Coefficients ($k_1 = 0.5, k_2 = 0.3$)

Under moderate revenue-sharing coefficients, the probability of brands selecting live streaming hosts and consumers choosing purchases gradually declined, eventually stabilizing at relatively low levels. Meanwhile, live streaming companies increasingly offer customized services, ultimately reaching a moderate equilibrium. This indicates that moderate revenue-sharing coefficients fail to fully satisfy the demands of both brands and consumers, resulting in a compromise equilibrium scenario where all three parties' returns remain below ideal levels.

5.3 Simulation Analysis of System Dynamic Evolution under Low Division Factor ($k_1 = 0.2, k_2 = 0.1$)

Under a low revenue-sharing coefficient, the probabilities that the brand chooses human streamers, the live streaming company provides customized services, and consumers opt to purchase all decreased gradually and eventually stabilized at a low level. This suggests that a low revenue-sharing coefficient fails to motivate any stakeholder to adopt strategies beneficial to the overall payoff, ultimately resulting in a compromised state with relatively low returns for all three parties.

6 Discussions

The numerical simulation results in this paper demonstrate that the evolution of the e-commerce live streaming industry is influenced by multiple factors, including the revenue-sharing coefficient, cost-benefit balance, consumer behavior, and technological innovation.

First, the revenue-sharing coefficient exerts a significant impact on evolutionary outcomes. A high revenue-sharing coefficient encourages brands and live streaming companies to prefer human streamers and customized services to pursue higher returns and a superior user experience, thereby forming a virtuous cycle that maximizes benefits for all participants. This finding aligns with the conclusions of Jiang et al. [13] and Fargetta and Scrimali [14] regarding the pivotal role of incentive mechanisms in live-streaming commerce, emphasizing that a reasonable benefit distribution is fundamental for coordinating multi-party strategies and achieving stable cooperation. Conversely, a low revenue-sharing coefficient drives brands and live streaming companies to opt for virtual streamers and standardized services to reduce costs; however, the payoffs for all parties fail to reach an ideal level. This is consistent with the cost-benefit analysis of service strategies by Zhou et al. [18], indicating that low-cost strategies are often associated with limited returns and user experience.

Second, the balance of costs and benefits directly affects the strategic choices of all stakeholders. Human streamers and customized services deliver higher returns and a better experience but incur greater costs. Virtual streamers and standardized services entail lower costs but are associated with relatively limited returns and user experience [18]. Therefore, selecting the appropriate streamer type and service model requires a comprehensive assessment by brands based on their positioning, target users, and market competition.

Furthermore, consumer behavior plays a vital role in the strategic choices of brands and live streaming companies. Consumer preferences regarding streamer types and service customization directly influence the decisions of all parties [19]. If consumers favor human streamers and customized services, brands and live streaming companies are more inclined to adopt such strategies; otherwise, corresponding adjustments are made. This corroborates the research of Zhao et al. [6] and Wan and Jiang [9], which identifies consumer trust-building and perception as key determinants of success in live-streaming e-commerce. Hence, understanding consumer behavior and implementing precision marketing are essential for the success of the e-commerce live streaming industry.

Finally, technological innovation serves as a crucial driving force for the development of the e-commerce live streaming industry. With the continuous advancement of virtual streamer technology and the expansion of application scenarios, the adoption of virtual streamers in e-commerce live streaming will become increasingly widespread [20]. Brands and live streaming companies should actively embrace new technologies and explore the application of virtual streamers to enhance user experience and reduce operational costs. Meanwhile, industry regulations should encourage innovation and provide support for the application of emerging technologies such as virtual streamers to promote the transformation and upgrading of the e-commerce live streaming industry. This trend is consistent with the research direction on virtual streamer interaction experiences by Chen and Li [3] and the overview of virtual human technology development by Cui and Liu [8].

7 Conclusions

This study uses numerical simulations based on an evolutionary game model to explore the evolutionary factors of the e-commerce live streaming industry. The results show that the industry's evolution is jointly driven by four key factors: revenue-sharing coefficient, cost-benefit balance, consumer behavior, and technological innovation. A high revenue-sharing coefficient facilitates a virtuous cycle of human streamers and customized services, while a low coefficient leads to cost-saving oriented choices of virtual streamers and standardized services. Cost-benefit trade-offs, consumer preferences, and technological progress also directly shape the strategic choices of stakeholders, and a reasonable revenue-sharing mechanism and technological innovation are crucial for the industry's healthy development.

8 Suggestions

Based on in-depth analysis of the evolution of the e-commerce live streaming industry, this paper proposed the following recommendations aimed at promoting the healthy development of the sector.

First, brands should formulate precise strategies. According to their own brand positioning and target consumer groups, brands should select the most suitable streamer type and customized services by integrating cost-benefit analysis. For instance, high-end brands may choose professional and personable human streamers and use customized services to shape a distinctive brand image. By contrast, mass-market brands may consider adopting more interactive and entertaining virtual streamers while controlling costs through standardized services. In addition, big data analytics could be employed to capture consumer behavior and preferences, so as to adjust strategies regarding streamer types and customized services and better meet consumer demands.

Second, live streaming companies should improve their service capabilities and technical expertise. They ought to provide diversified streamer options and customized services to meet the requirements of different brands. For example, they can offer human streamers of various types and styles, as well as virtual streamers with multiple functions and distinctive features. Meanwhile, they should strengthen research and development (R&D) in virtual streamer technology to enhance their realism and interactivity, in order to enable better engagement with consumers and deliver a personalized shopping experience. This may include developing highly realistic virtual streamer images or intelligent interactive systems.

Finally, consumers should participate actively in interactions and make rational choices of products. When selecting streamer types and customized services, consumers should make rational decisions based on their personal needs and preferences. For instance, consumers who value product authenticity and credibility may choose products promoted by human streamers, while those who prioritize entertainment and interactivity may prefer products introduced by virtual streamers. Meanwhile, actively engaging in live streaming interactions and providing feedback can contribute to the healthy development of the e-commerce live streaming industry. In addition, consumers should pay attention to the authenticity of streamers and products to avoid being misled by false publicity and protect their legitimate rights and interests. This includes verifying streamers' qualifications and product reviews, or reporting false advertising practices to relevant authorities.

Through the aforementioned recommendations, collaborative efforts by brand owners, live streaming companies, and consumers could foster the healthy growth of the e-commerce live streaming industry, thus achieving a win-win scenario.

Author Contributions

Conceptualization, J.Y.Z. and R.Q.Y.; methodology, J.Y.Z.; software, R.Q.Y.; validation, J.Y.Z.; formal analysis, J.Y.Z.; investigation, J.Y.Z. and R.Q.Y.; resources, J.Y.Z. and R.Q.Y.; data curation, R.Q.Y.; writing—original draft preparation, J.Y.Z. and R.Q.Y.; writing—review and editing, J.Y.Z.; visualization, J.Y.Z. and R.Q.Y.; supervision, J.Y.Z. and R.Q.Y.; project administration, J.Y.Z. and R.Q.Y.; funding acquisition, J.Y.Z. and R.Q.Y. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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